Duplicate images and leaking splits in public polyp segmentation benchmarks

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Abstract

Automated polyp segmentation is benchmarked almost exclusively on a small set of public datasets distributed together as one evaluation suite. We audit those artifacts at the image level. All 60 CVC-300 images are pixel-identical to CVC-ColonDB frames, so the two are not independent test sets; 34 of 380 CVC-ColonDB files are exact duplicates of other files in the same folder; and the distributed CVC-ColonDB cannot be reconciled with its published specification of 300 frames selected to maximise visual difference. Correcting the double counting shifts every architecture's mean external Dice by 2.6–4.8 times the protocol's measured run-to-run noise floor and reverses which architecture ranks first. Frames are redundant enough that every random split of CVC-ClinicDB leaks, inflating in-domain Dice by 0.0970 (18.3σ), whereas Kvasir-SEG inflates by only 0.0059. We release leakage-free splits, recovered sequence labels and the audit code.

Introduction

Progress in automated polyp segmentation is reported almost entirely through a single evaluation template. A model is trained on Kvasir-SEG and CVC-ClinicDB, then scored on CVC-ColonDB, ETIS-LaribPolypDB and CVC-300, and the five resulting Dice values are compared against a leaderboard inherited from PraNet (Fan et al., 2020). Recent work has begun to ask what that template can establish. Reported differences between architectures can fall below the run-to-run variance of the training procedure itself (Puri, 2026a), split protocols that co-exist in the literature are mutually incomparable, and near-tied rankings reverse when the partition is redrawn (Urooj et al., 2026). General guidance on validation practice has been issued for medical image analysis as a whole (Maier-Hein et al., 2024), but it presumes that the datasets under evaluation are what their documentation says they are.

Those critiques treat the datasets as given and ask whether the statistics computed on them are sound. We ask the prior question of whether the artifacts themselves are what they are believed to be. This is not a hypothetical concern: duplicate and near-duplicate images have been shown to contaminate widely used dermatology benchmarks and to change the conclusions drawn from them (Cassidy et al., 2022; Gröger et al., 2023), and the colonoscopy datasets have a structural reason to be vulnerable. Every one of them is assembled from frames extracted from a small number of colonoscopy videos, so visually near-identical frames are available in abundance, and most of the datasets never published a mapping from frame to source video.

We audit the suite at the level of individual images and report three classes of finding. First, the suite contains redundancy that is invisible to the protocol: one test set is contained in another,

Table 1: **Image-level audit of the evaluation suite.** Files counts the images as distributed; Unique counts distinct images by the MD5 digest of the decoded pixels. Kvasir-Sessile is distributed separately and its images are drawn from Kvasir-SEG, so it is excluded from suite totals.

Dataset	Files	Unique	Exact	Near-duplicate (%)		Contained in
			dupl.	$r > 0.95$	$r > 0.90$	
Kvasir-SEG	1000	1000	0	1.2	15.3	–
Kvasir-Sessile	196	196	0	2.0	12.2	Kvasir-SEG
CVC-ClinicDB	612	612	0	57.5	90.0	–
CVC-ColonDB	380	346	34	67.4	85.8	–
CVC-300	60	58	2	80.0	83.3	CVC-ColonDB
ETIS-Larib	196	193	3	56.6	77.5	–

several files are exact duplicates, and the proportion of images carrying a near-duplicate reaches 90.0% in CVC-ClinicDB against 15.3% in Kvasir-SEG. Second, the circulating CVC-ColonDB cannot be reconciled with the dataset its creators describe, and the authoritative copy needed to reconcile them is no longer obtainable. Third, these defects are consequential rather than cosmetic: they move reported scores and architecture rankings by multiples of the noise floor that the same protocol exhibits under repeated identical training.

Throughout we report effects in units of $\sigma_{\text{run}} = 0.0053$ Dice, the run-to-run noise floor measured for this training recipe (Puri, 2026a,b), because an effect that cannot be distinguished from rerunning the same configuration cannot support a claim. We release leakage-free splits, recovered sequence labels, per-image results for all 100 training runs, and the audit code.

Results

One external test set is contained in another

Table 1 summarises the audit and Table 2 the consequences measured from it.

Across the 2,248 images of the suite, every one of the 60 CVC-300 images is pixel-identical to an image in CVC-ColonDB, occupying the contiguous frame range 149–208 (Fig. 1a). Agreement holds under two independent criteria, MD5 of the decoded pixel array and a 256-bit perceptual hash at zero Hamming distance. A third, fully independent check confirms it downstream: scored by five trained architectures, CVC-300 and CVC-ColonDB frames 149–208 yield mean Dice agreeing to within 1.9e-06.

Consequently CVC-300 and CVC-ColonDB are not independent test sets, and averaging them as separate datasets counts those 60 images twice. The duplicated block is also unrepresentative of its parent: it scores 0.8534 against 0.7216 for the remainder of CVC-ColonDB, a gap of +0.1317 Dice. Reporting it separately therefore counts an unusually easy subset twice.

Exact duplicates and pervasive near-duplication

CVC-ColonDB contains 34 files that are byte-level duplicates of other files in the same folder, leaving 346 unique images among 380; ETIS-LaribPolypDB contains 3. Beyond exact copies, the proportion of images possessing a near-duplicate within the same dataset is 90.0% for CVC-ClinicDB, 85.8% for CVC-ColonDB and 77.5% for ETIS-LaribPolypDB, against 15.3% for Kvasir-SEG (Fig. 1b).

The audit also returns negative results that bound its scope. No image is shared between Kvasir-SEG and any CVC dataset, CVC-ClinicDB and CVC-ColonDB share no images despite

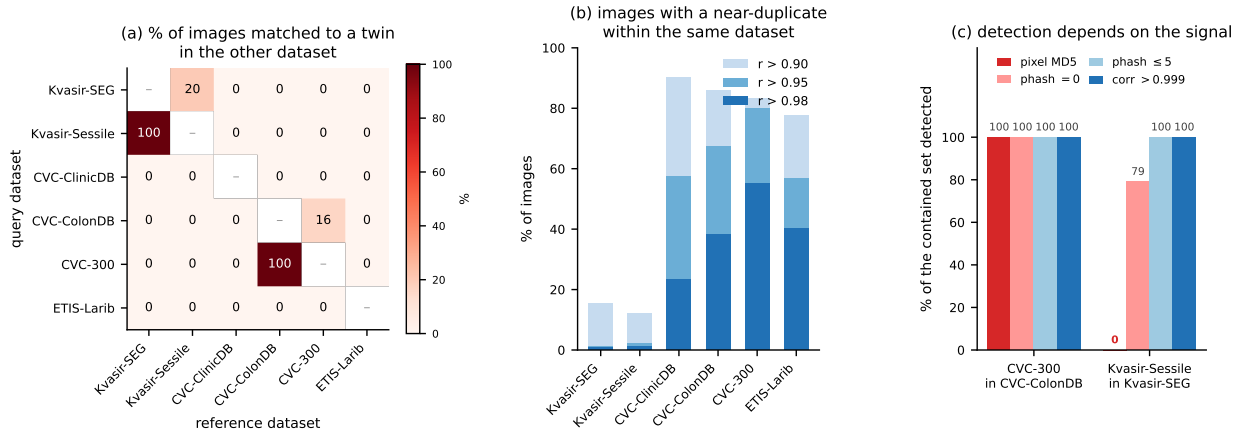


Figure 1: **What the benchmark suite contains.** (a) Percentage of each dataset's images matched to a twin in another dataset, under the criterion that the correlation and perceptual-hash signals agree. (b) Percentage having a near-duplicate within the same dataset at three correlation thresholds. (c) Proportion of each contained set recovered by four detection signals; exact pixel hashing recovers none of Kvasir-Sessile.

common provenance, and ETIS-LaribPolypDB is clean against every other set.

A second containment is invisible to exact hashing

Kvasir-Sessile is distributed separately from Kvasir-SEG and is sometimes treated as a held-out set of flat and sessile lesions. All 196 of its images correspond to a Kvasir-SEG image: they share filenames, have identical dimensions, and match at a thumbnail correlation above 0.999 in 196 of 196 cases and at a perceptual-hash distance of at most five bits in 196. Kvasir-Sessile is therefore contained in Kvasir-SEG, and using it as a test set for a model trained on Kvasir-SEG admits the whole of it into training.

The two containments are nonetheless detected very differently (Fig. 1c). CVC-300 is recovered completely by every signal, including the MD5 digest of the decoded pixels. Kvasir-Sessile is recovered completely by correlation and by perceptual hashing at a tolerance of five bits, by 155 of 196 at zero perceptual-hash distance, and by 0 of 196 by exact pixel digest, because the images were re-encoded between the two releases. An audit resting on checksums alone would report this dataset as clean. Duplicate detection in benchmark suites therefore cannot rely on exact hashing, a point that applies well beyond the datasets examined here.

The distributed CVC-ColonDB cannot be reconciled with its specification

The CVC-ColonDB that circulates in the evaluation suite differs from the dataset its creators describe. The authoritative project page specifies 15 short colonoscopy sequences from 15 studies, with region-of-interest annotations provided for 300 frames, distributed as 1200 files comprising the originals and three mask types. Those 300 frames are stated to have been "selected in order to maximise the visual difference between them".

The copy distributed with the benchmark suite contains 380 image files, of which 346 are unique. Neither figure matches the specified 300, and the surplus cannot be characterised: the download host specified by the creators has been unreachable, the page itself has not been revised since 2018, and neither the archive nor the registration endpoint was ever captured by the Internet Archive, so

Table 2: **Measured consequences, in Dice and in units of the run-to-run noise floor** $\sigma_{\text{run}} = 0.0053$. Negative values indicate that the corrected figure is lower than the reported one. The position prior is an absolute score rather than a difference, so no noise-floor ratio is given.

Effect	Dice	$/\sigma_{\text{run}}$	Note
Dropping CVC-300 from the suite average (5 architectures)	-0.0214	4.0	top rank reverses; 2/5 move
Dropping CVC-300, released PraNet checkpoint	-0.0669	12.6	three unseen datasets become two
Random vs sequence-disjoint splits, CVC-ClinicDB	+0.0970	18.3	612 frames × 5 arch., paired, $p \approx 10^{-339}$
adjusted for training diversity	+0.0701	13.2	estimate, see text
Random vs grouped splits, Kvasir-SEG	+0.0059	1.1	3/5 intervals cross zero; $p \approx 10^{-6}$
Content-free position prior, Kvasir-SEG	0.3963	–	vs 0.8996 trained
Content-free position prior, CVC-ClinicDB	0.2982	–	vs 0.8231 trained

no authoritative copy remains against which to reconcile the difference. We therefore report the discrepancy without attributing a cause.

The curation statement is nonetheless directly testable against the circulating artifact, and it does not hold: 85.8% of the distributed images carry a near-duplicate at $r > 0.90$ and 34 are pixel-identical copies. Whatever the 380-file artifact is, it is not a set of frames selected to maximise visual difference. Because the authoritative copy is unobtainable, this is also a plausible explanation for how the discrepancy has persisted unremarked: there has been nothing to check against.

The containment affects the model that defined the protocol

The results so far concern architectures we trained ourselves. Because PraNet released both the protocol and its trained weights, the containment can be examined on the model the field actually cites, with no retraining and without our training recipe entering the comparison. Running the released checkpoint over the same images reproduces the published figures closely, giving 0.7153 on CVC-ColonDB, 0.6367 on ETIS-LaribPolypDB and 0.8767 on CVC-300 against the 0.709, 0.628 and 0.871 reported, which indicates that the evaluation pipeline used here is faithful.

On that checkpoint, CVC-300 and CVC-ColonDB frames 149–208 yield mean Dice differing by 1.9e-06, and the duplicated block scores 0.8767 against 0.6851 for the remainder of CVC-ColonDB, a gap of +0.1916 Dice. The protocol presents three unseen datasets, but two of them are the same images, and the easier of the two subsets is counted twice. Averaging the three as reported gives 0.7429; averaging the two that are genuinely distinct gives 0.6760, a difference of -0.0669 Dice, or $12.6\sigma_{\text{run}}$. The claim supported by this protocol is generalisation to two unseen datasets rather than three.

CVC-ClinicDB is excluded from that average because the checkpoint was trained on 550 of its 612 images, so it is not a held-out set for this model. Evaluating it on all 612 regardless returns 0.9697 against the 0.899 reported for the 62 images actually held out, which indicates the size of the gap that scoring a model on its own training data can produce.

Correcting the double counting reverses the leading architecture

Recomputing the benchmark from per-image metrics for five architectures under five-fold cross-validation, with CVC-300 removed as a duplicate of CVC-ColonDB and exact duplicates collapsed,

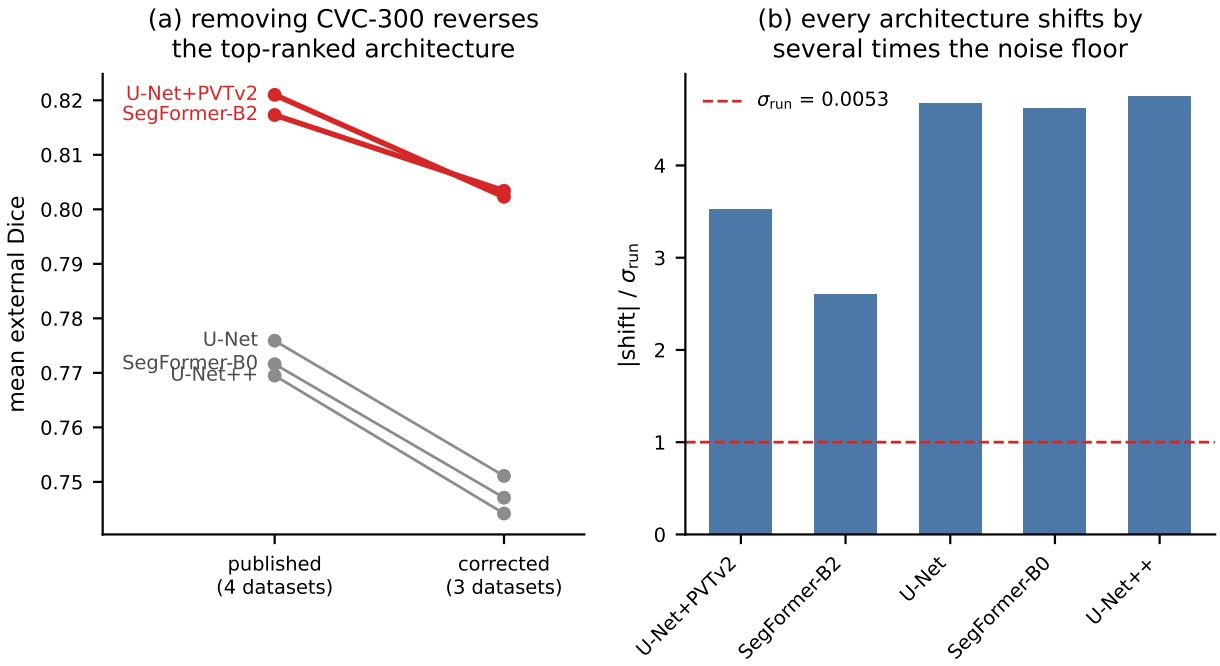


Figure 2: **Consequence of counting CVC-300 as an independent test set.** (a) Mean external Dice before and after removing the duplication; the architectures that exchange rank are highlighted. (b) Magnitude of the shift in units of σ_{run}.

shifts mean external Dice by 0.0138–0.0252 Dice, that is 2.6–4.8σ_{run}, for 5 of 5 architectures. The ordering changes: unet_pvt ranks first under the published four-dataset average and segformer_b2 ranks first once the duplication is removed (Fig. 2). 1 of 10 pairwise comparisons change sign, and it is the comparison that decides the leading architecture.

Leaking splits inflate CVC-ClinicDB by eighteen noise floors

Near-duplication within a training set has a direct consequence: a random assignment of frames to folds places visually near-identical frames on both sides of the partition. Using the official frame-range table, which assigns each of CVC-ClinicDB's 612 frames to one of 29 source sequences, every one of 1000 random 80/20 partitions placed frames from all of the test sequences in the training set as well.

We trained five architectures over five folds under one recipe twice: once with frames assigned at random, as the literature does, and once with whole source sequences held out. Both arms partition all 612 frames, so every frame is held out exactly once per protocol and the two arms can be compared on identical images rather than on different test folds, giving 3,060 paired observations across the five architectures. Random splits yield a mean Dice of 0.9201 against 0.8231 for sequence-disjoint splits, an inflation of 0.0970 Dice [0.0911, 0.1031], or 18.3σ_{run}. The effect is present in every architecture (0.0800–0.1180 Dice, 15.1–22.3σ) and is far larger than the spread between architectures (Fig. 3a).

Repeating the comparison on Kvasir-SEG, whose near-duplicate rate is 15.3% rather than 90.0%, yields an inflation of 0.0059 Dice, or 1.1σ_{run}, with intervals crossing zero for three of five architectures (Fig. 3b). The magnitude of the effect therefore tracks the redundancy of the dataset rather than the choice of partitioning procedure in itself.

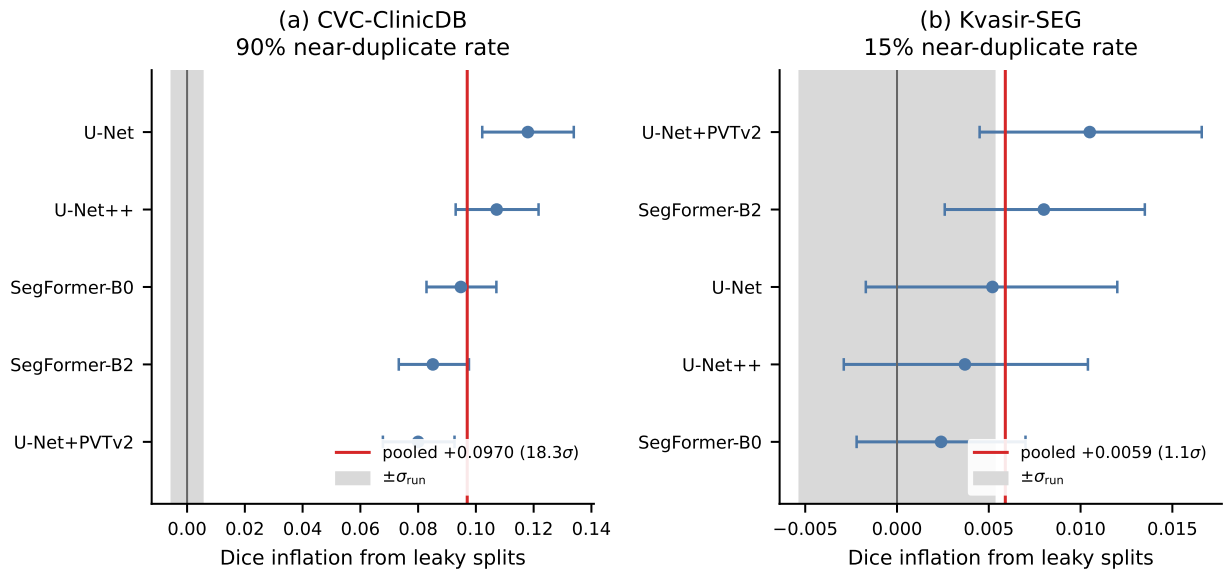


Figure 3: **Inflation attributable to leaking splits.** Per-architecture inflation with percentile bootstrap intervals, pooled estimate in red and the $\pm\sigma_{\text{run}}$ band shaded, for (a) CVC-ClinicDB and (b) Kvasir-SEG.

Part of the in-domain difference is not leakage. Scores on the external test sets, which neither arm trained on, also differ between the arms by 0.0198–0.0382 Dice. Leakage cannot act on data that was never seen, and the two arms differ in a second respect: at equal training-set size, the sequence-disjoint folds exclude roughly a fifth of the source sequences and so present fewer distinct lesions. Taking the mean external difference 0.0269 as an estimate of that contribution gives an adjusted in-domain estimate of 0.0701 Dice, or $13.2\sigma_{\text{run}}$, attributable to leakage proper. The corresponding external differences in the Kvasir-SEG experiment, where grouping removes almost no lesion diversity, are 0.0027–0.0183 Dice, which is consistent with this account.

Unlike the duplication of CVC-300, leakage did not change the in-domain ordering of architectures on CVC-ClinicDB, where 0 of 5 architectures moved. On Kvasir-SEG, where the mean effect is at the noise floor, 2 of 5 moved, because the two leading architectures are separated by less than σ_{run}. A shift in the mean and a change in the ranking are distinct failures, and neither implies the other.

A predictor that never sees the image

Benchmark polyps are typically centred and well framed, a bias previously reported as a dataset statistic (Ji et al., 2022). To express that bias in the metric the field reports, we constructed a predictor that sees no image content: it averages the training-fold masks into a single probability map, thresholds it at the value maximising training Dice, and emits that one fixed shape for every test image.

The prior alone reaches 0.3963 Dice on Kvasir-SEG and 0.2982 on CVC-ClinicDB, against 0.8996 and 0.8231 for models trained under the leak-free protocol (Fig. 4a). Values across the four datasets are ordered exactly by mean lesion area (Fig. 4b), which identifies the mechanism: a single fixed blob overlaps large, centrally framed lesions and misses small or peripheral ones. Dice is not linear in task difficulty, so these figures should not be read as a fraction of the problem solved. They establish a floor below which a reported score carries no evidence of lesion recognition at all, and on

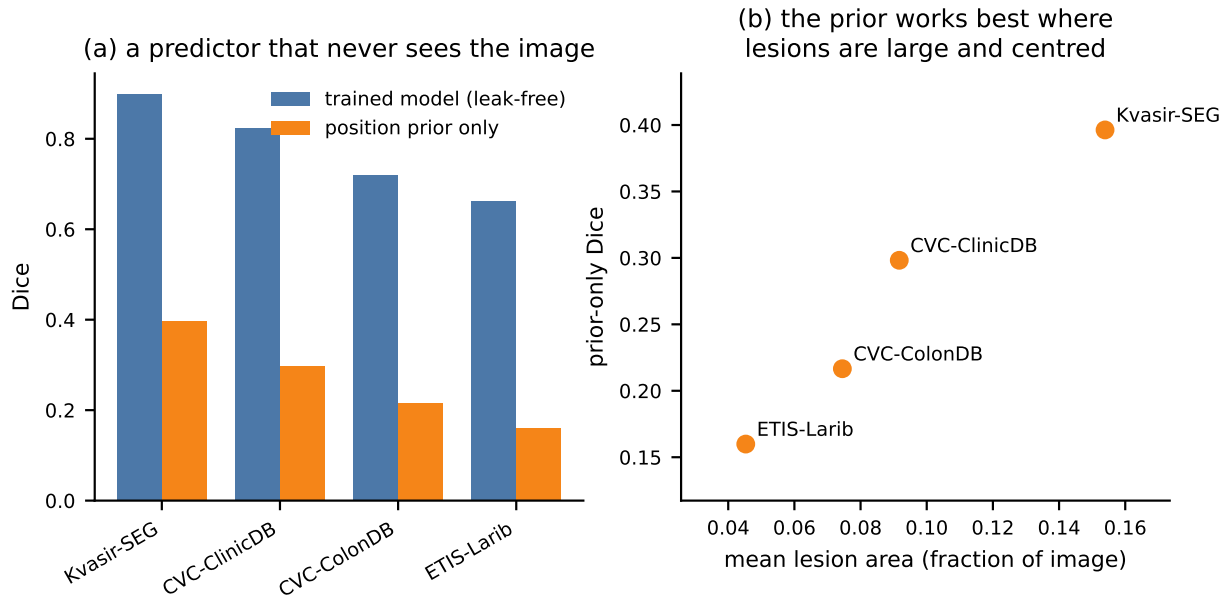


Figure 4: **Dice obtainable without seeing the image.** (a) Trained models against a content-free position prior. (b) Prior-only Dice against mean lesion area.

the two datasets used for training and in-domain reporting that floor is near 0.2982–0.3963.

Source video structure is recoverable from pixels

Leakage-free partitioning requires knowing which frames share a source video, and most of these datasets never published that mapping. It is nonetheless recoverable. Because frames from one sequence occupy a contiguous range of the distributed numbering, recovery is a segmentation problem on an ordered stream rather than a clustering problem. Partitioning CVC-ClinicDB into contiguous segments that minimise within-segment feature variance recovers the 29 official sequences at an adjusted Rand index of 0.997, with boundaries located to within two frames at an F1 of 1.000 (Fig. 5). Treating the same problem as unordered clustering reaches only 0.497, and thresholding a similarity graph fails outright because the graph percolates into a single component.

Exact recovery is not in fact required. The two error directions are not symmetric: splitting one true sequence across groups permits leakage, whereas merging two sequences into one group does not, costing only partitioning granularity. The quantity to control is therefore completeness rather than agreement, and a deliberately coarse segmentation satisfies it. On CVC-ClinicDB no true sequence is split for any segment count up to $K/n = 0.041$, at which point the grouping still attains a homogeneity of 0.953. Estimating the number of sequences without ground truth proved unreliable, so we certify this property only where a published table exists, and treat Kvasir-SEG, which carries no frame ordering at all, as uncertified.

Discussion

These findings concern the validity of a measurement instrument, not the competence of the groups who built the datasets or the correctness of any particular published number. The distinction is worth stating precisely, because the two kinds of claim carry very different evidential burdens.

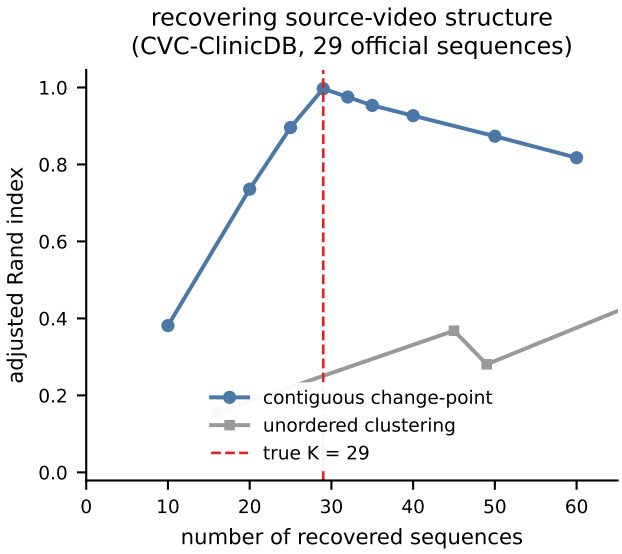


Figure 5: **Recovering source video structure.** Agreement with CVC-ClinicDB's 29 official sequences for contiguous change-point segmentation and for unordered clustering.

That CVC-300 is contained in CVC-ColonDB is not a statistical inference. The images are identical under MD5 of their decoded pixel arrays and under a 256-bit perceptual hash at zero Hamming distance, and the two sets produce mean Dice agreeing to within 1.9e-06 when scored by five independently trained architectures. Anyone can reproduce the check in minutes. From that identity, one consequence follows deductively and requires no experiment at all: a mean taken over CVC-300 and CVC-ColonDB as though they were separate datasets counts 60 images twice, and because that block is +0.1317 Dice easier than the remainder of its parent, the double counting is also biased upward rather than neutral. The same holds for the 34 exact duplicates. These are properties of the artifacts, and they hold for every study that has used them.

The magnitude of the leakage effect is a weaker claim and we state it as one. It was measured for five architectures spanning four backbone families under a single training recipe, and we did not retrain any published model. We therefore cannot say what any specific reported score would become under a leakage-free protocol. What makes extrapolation reasonable is that the mechanism resides in the data and the partition rather than in the architecture: a test frame whose sequence-mates appear in training is easier for any model. That the effect appeared in all five architectures, and that its size tracked dataset redundancy rather than the partitioning procedure, supports that reading, but it remains an inference. The decomposition is likewise approximate; using external performance to estimate the training-diversity contribution assumes that contribution transfers equally in and out of domain, which we cannot verify directly.

Our own prior work is subject to the same audit and we report its exposure rather than exempting it. That study used the distributed CVC-ColonDB as an external test set and cross-validated Kvasir-SEG at the image level, listing the absence of patient identifiers as a limitation. The present measurement puts that exposure at 0.0059 Dice, or $1.1\sigma_{\text{run}}$, which leaves its conclusions intact but converts an acknowledged caveat into a quantity.

Two of our own methods failed and the failures are informative. Thresholding a similarity graph and taking connected components, the obvious way to group near-duplicates, percolates: by a correlation of 0.77 all 612 CVC-ClinicDB frames collapse into one component, at which point

the grouping is trivially leak-safe and useless. Selecting the number of source sequences without ground truth was also unstable, with a slope heuristic recovering 32 against a true 29 on the full dataset but 27 and 25 against a true 15 on its halves. We consequently certify leak-safety only for CVC-ClinicDB, where a published table exists, and offer the recovered groupings for the remaining datasets as a mitigation rather than a guarantee. A third limitation is that the near-duplicate rates depend on a correlation threshold; the exact-duplicate and containment results do not, which is why the argument rests on them.

The scope of the audit is also deliberately narrow. We examine the specific bundle of artifacts that the prevailing protocol distributes, because that is what published scores are computed on, and not the newer resources that have since become available. Multi-centre collections such as PolypGen (Ali et al., 2023) and full-procedure video datasets such as REAL-Colon (Biffi et al., 2024) were assembled with provenance metadata and centre or patient identifiers, which is precisely what the older datasets lack, and they are better positioned to support the kind of partitioning we recommend. Whether they contain comparable defects is an open question that the released toolkit is intended to make cheap to answer; nothing in our results speaks to it either way.

What follows for practice is modest and cheap. CVC-300 should not be reported as a dataset independent of CVC-ColonDB, and an average over the suite should not include both. Exact duplicates should be collapsed before scoring. Splits should respect source sequences wherever a mapping exists or can be recovered, and where it cannot, the resulting uncertainty should be stated rather than assumed away. Effects smaller than a measured run-to-run noise floor should not be interpreted, a point established previously (Puri, 2026a) and reinforced here, since the corrections we describe are several times that floor while many published differences are not. To make this inexpensive we release leakage-free splits, the recovered sequence labels, the contamination matrix, per-image metrics for all 100 runs, and the audit code, so that the checks reported here can be rerun on any new dataset before it enters general use.

Methods

Datasets and the distributed evaluation suite

The audit covers the 2,248 images of the five datasets that constitute the standard suite: Kvasir-SEG (Jha et al., 2020) (1000), CVC-ClinicDB (Bernal et al., 2015) (612), CVC-ColonDB (Bernal et al., 2012) (380), ETIS-LaribPolypDB (Silva et al., 2014) (196) and CVC-300, the test portion of EndoScene (Vázquez et al., 2017) (60). We additionally audit Kvasir-Sessile (196), which is distributed separately; it is excluded from the suite total because, as reported in the Results, its images are drawn from Kvasir-SEG and counting them there would double-count them. The last three were taken from the test bundle distributed with PraNet (Fan et al., 2020), which is the artifact the literature actually evaluates on; we record its SHA-256 in the deposit so that the specific copy audited here is identifiable. CVC-ClinicDB was obtained in its full 612-image form rather than the 62-image subset in that bundle, which preserves the original frame numbering and hence the official sequence table. Image data are not redistributable and are not included in our release; Kvasir-SEG in particular is licensed for research and educational use only.

Image-level duplicate detection

Two deliberately dissimilar signals were computed per image so that neither alone drives a conclusion, a precaution that proved necessary: one of the two containments we report is invisible to exact hashing. Exact identity used the MD5 digest of the decoded RGB pixel array, which is invariant to

container format and file metadata but not to any change in pixel values, including lossy re-encoding between dataset releases. Near-duplicate similarity used 64×64 mean-subtracted, L2-normalised greyscale thumbnails compared by Pearson correlation, which is invariant to resolution, together with a 256-bit perceptual hash compared by Hamming distance, which tolerates mild cropping and recompression. A pair was treated as the same frame only when both criteria agreed, and all candidate matches above the reporting threshold were written out with both file paths for inspection. Near-duplicate rates are reported at $r > 0.90$, 0.95 and 0.98 because the rate is threshold-dependent; the exact-duplicate and containment results involve no threshold.

Recovering source video structure

Frames belonging to one source sequence occupy a contiguous range of the distributed numbering, so recovery was posed as segmentation of an ordered stream. Each frame was represented by its normalised greyscale thumbnail concatenated with per-channel colour histograms, each block L2-normalised so that neither dominates. The frame order was partitioned into exactly K contiguous segments minimising the total within-segment sum of squared deviations, solved exactly by dynamic programming with prefix sums rather than greedily. CVC-ClinicDB provides ground truth through its official frame-range table of 29 sequences, against which recovery was scored by adjusted Rand index, homogeneity, completeness and boundary F1 at tolerances of zero, one and two frames. Unordered agglomerative clustering on the same features, and connected components of a thresholded similarity graph, were evaluated as alternatives; the latter percolates and was discarded. Leak-safety was defined as completeness equal to one, that is no true sequence split across groups, which an intentionally coarse segmentation attains without requiring K to be known.

Training protocol

Five architectures spanning four backbone families were trained under one recipe identical to that of ref. (Puri, 2026a): U-Net and U-Net++ with ResNet-34 encoders, SegFormer-B0 and SegFormer-B2, and U-Net with a PVTv2-B2 encoder. Inputs were resized to 352×352 and optimised with AdamW (learning rate 10^{-4}, weight decay 10^{-4}) under a cosine schedule, batch size 16, at most 60 epochs with early stopping on validation Dice at a patience of 15, minimising the sum of Dice and binary cross-entropy losses under bfloat16 autocast. Predictions were resized to native resolution before scoring. The released code of ref. (Puri, 2026a) resolves image and mask paths under the Kvasir-SEG naming convention only; rather than modify it, which would weaken the claim that both studies share one recipe, we widened the record builder through an adapter that honours explicit filenames when a split file supplies them and otherwise delegates to the original function. All 100 runs were executed on a single RTX 4090.

Paired comparison of split protocols

Comparing the two protocols by their reported means would be confounded, since the protocols necessarily induce different test folds and a difference would mix leakage with the test sets containing different images. Both arms were therefore constructed as five-fold partitions of the complete dataset, so that every frame is held out exactly once under each protocol and the arms can be compared image by image. Validation data were carved from whole groups within the training portion so that the sequence-disjoint property holds for validation as well. Inflation was summarised by the mean per-image difference with percentile bootstrap intervals over 10^4 resamples, and tested by paired Wilcoxon signed-rank. Training-set sizes were matched as closely as whole-sequence assignment

permits (416–430 against 428–429 frames) so that data volume is not a confound. All effects are reported both in Dice and in units of $\sigma_{\text{run}} = 0.0053$.

Statistics and reproducibility

No sample size calculation was performed; the analysis uses every image in each dataset, so sample sizes are fixed by the data rather than chosen. No data were excluded except where stated, namely the collapsing of exact duplicates and the removal of CVC-300 as a duplicate of CVC-ColonDB, both of which are the subject of the study rather than data cleaning. Allocation of images to folds was at random within the constraints described above; blinding is not applicable, as all outcomes are computed automatically from stored predictions with no human scoring.

Comparisons between split protocols are paired at the level of the individual image, each image being held out exactly once under each protocol, and are tested with the two-sided Wilcoxon signed-rank test. A rank-based paired test was chosen because per-image Dice is bounded on $[0, 1]$ and strongly left-skewed, so the differences are not normally distributed and a paired t-test would be inappropriate; no normality assumption is required for the Wilcoxon test. The in-domain comparison on CVC-ClinicDB uses $n = 3,060$ paired observations (612 images $\times$ 5 architectures), of which 3,059 differ, and gives $W = 414,915$, $z = -39.4$ and a two-sided $p \approx 10^{-339}$. The corresponding comparison on Kvasir-SEG uses $n = 5,000$ paired observations and gives $W = 5,753,106$, $z = -4.8$ and $p = 1.89 \times 10^{-6}$. Because the exact two-sided p-value underflows double precision at these sample sizes, p is obtained from the standardised statistic of the normal approximation, for which $\log_{10} p$ remains finite; the exact routine returns zero and is not reported as such.

The alpha level is 0.05 throughout, though no conclusion in this work rests on a threshold: effects are interpreted against the measured run-to-run noise floor $\sigma_{\text{run}} = 0.0053$ Dice rather than against statistical significance, since with thousands of paired images a difference far below that floor would still be detected as significant. Uncertainty on every effect is reported as a percentile bootstrap interval over 10^4 resamples of the per-image differences. All training runs used a fixed seed, and the released per-image metrics allow every reported statistic to be recomputed without retraining.

Content-free position prior

For each fold, the training-fold masks were averaged on a 352×352 grid into a single probability map and binarised at the threshold maximising mean Dice on the training fold. That one fixed shape was then scored against every held-out mask. The predictor receives no image as input, so any Dice it attains is attributable to the dataset's position and size prior alone. Folds were the leak-free ones where a defensible grouping exists, and a plain partition otherwise, which is noted in the results.

Data availability

All derived data generated by this study are deposited at `https://doi.org/10.5281/zenodo.23879374`: the full image index with per-image pixel digests and perceptual hashes, the cross-dataset contamination matrix and the list of matched pairs, the exact-duplicate groups, within-dataset near-duplicate rates, leakage-free split and fold definitions, recovered sequence labels for CVC-ClinicDB with the official labels alongside for verification, deduplicated test-set keep-lists, and per-image segmentation metrics for all 100 training runs. The archived specification of CVC-ColonDB referred to in the Results is available through the Internet Archive capture of the creators' project page.

The underlying colonoscopy images are not redistributable and are not included. They must be obtained from their original providers: Kvasir-SEG and Kvasir-Sessile from the Simula datasets portal, CVC-ClinicDB and ETIS-LaribPolypDB through their respective challenge sites, and CVC-ColonDB and CVC-300 as distributed with PraNet. Kvasir-SEG is licensed for research and educational purposes only. Every row of the released metrics identifies its source image by filename so that results can be joined back to the original data.

Code availability

The audit toolkit and all analysis code are available at `https://github.com/rishitpuri/polyp-benchmark-audit` and archived at `https://doi.org/10.5281/zenodo.23879374`, including the duplicate-detection and sequence-recovery implementations, the split generators, the analysis scripts that produce every reported number, and the figure scripts. Every value quoted in this manuscript is generated from the released result files rather than transcribed, so the text cannot diverge from the experiments.

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Author contributions

R.P. is the sole author. R.P. conceived and designed the study, implemented the audit and analysis software, performed all experiments, analysed and interpreted the results, produced the figures and tables, prepared the data and code release, and wrote the manuscript.

Competing interests

The author is employed by Karl Storz Endoscopy-America, Inc. This study was conceived and conducted independently of that employment. It uses only publicly available datasets; no proprietary, employer-held or patient data were used, and no employer resources, funding or personnel contributed to the design, analysis, or reporting of the work. The employer had no role in the decision to publish. The author declares no other competing interests.

References

Deng-Ping Fan, Ge-Peng Ji, Tao Zhou, Geng Chen, Huazhu Fu, Jianbing Shen, and Ling Shao. PraNet: Parallel reverse attention network for polyp segmentation. In *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, pages 263–273, 2020. arXiv:2006.11392.

Rishit Puri. What polyp segmentation benchmarks cannot measure: small lesions, domain shift, and a noise floor larger than most reported gains. *Under review*, 2026a.

Aisha Urooj, Zain Ul Abdien, and Neelu Madan. Metrics or mirage? an audit of evaluation inconsistencies in colonoscopy polyp segmentation benchmarks. *arXiv:2607.08203*, 2026.

Lena Maier-Hein, Annika Reinke, Patrick Godau, et al. Metrics reloaded: recommendations for image analysis validation. *Nature Methods*, 21:195–212, 2024.

Bill Cassidy, Connah Kendrick, Andrzej Brodzicki, Joanna Jaworek-Korjakowska, and Moi Hoon Yap. Analysis of the ISIC image datasets: usage, benchmarks and recommendations. *Medical Image Analysis*, 75:102305, 2022.

Fabian Gröger, Simone Lionetti, Philippe Gottfrois, Alvaro Gonzalez-Jimenez, Ludovic Amruthalingam, et al. Towards reliable dermatology evaluation benchmarks. In *Machine Learning for Health (ML4H), PMLR*, volume 225, 2023.

Rishit Puri. Per-image evaluation metrics for 88 polyp segmentation training runs, 2026b. Companion data release.

Ge-Peng Ji, Guobao Xiao, Yu-Cheng Chou, Deng-Ping Fan, Kai Zhao, Geng Chen, and Luc Van Gool. Video polyp segmentation: a deep learning perspective. *Machine Intelligence Research*, 19:531–549, 2022.

Sharib Ali, Debesh Jha, Noha Ghatwary, et al. A multi-centre polyp detection and segmentation dataset for generalisability assessment. *Scientific Data*, 10:75, 2023.

Carlo Biffi, Giulio Antonelli, Sebastian Bernhofer, et al. REAL-Colon: A dataset for developing real-world AI applications in colonoscopy. *Scientific Data*, 11:539, 2024.

Debesh Jha, Pia H Smedsrud, Michael A Riegler, Pål Halvorsen, Thomas de Lange, Dag Johansen, and Håvard D Johansen. Kvasir-SEG: A segmented polyp dataset. In *MultiMedia Modeling (MMM)*, pages 451–462, 2020.

Jorge Bernal, F Javier Sánchez, Gloria Fernández-Esparrach, Debora Gil, Cristina Rodríguez, and Fernando Vilariño. WM-DOVA maps for accurate polyp highlighting in colonoscopy: validation vs. saliency maps from physicians. *Computerized Medical Imaging and Graphics*, 43:99–111, 2015.

Jorge Bernal, Javier Sánchez, and Fernando Vilariño. Towards automatic polyp detection with a polyp appearance model. *Pattern Recognition*, 45(9):3166–3182, 2012.

Juan Silva, Aymeric Histace, Olivier Romain, Xavier Dray, and Bertrand Granado. Toward embedded detection of polyps in WCE images for early diagnosis of colorectal cancer. *International Journal of Computer Assisted Radiology and Surgery*, 9(2):283–293, 2014.

David Vázquez, Jorge Bernal, F Javier Sánchez, Gloria Fernández-Esparrach, Antonio M López, Adriana Romero, Michal Drozdzal, and Aaron Courville. A benchmark for endoluminal scene segmentation of colonoscopy images. *Journal of Healthcare Engineering*, 2017:4037190, 2017.